

Task 1. Which normal form is violated

1NF requires every cell to hold one atomic value. `seat_numbers` stores a list like "12A, 12B", so one cell contains several values. Because of this we can't easily query who sits in seat 12B. The same problem exists in `passenger_full_name` and `passenger_passport_number` when one booking has several passengers.

2NF requires every non-key column to depend on the whole primary key. After fixing 1NF the key becomes (`booking_id`, `passenger_passport_number`). But `passenger_full_name` depends only on the passport number, and `flight_number` and `ticket_price` depend only on `booking_id`. These are partial dependencies.

3NF requires that non-key columns don't depend on other non-key columns. Here `flight_number` determines `airline_name` and the airports, and `airport_name` determines `city`. These are transitive dependencies. They cause redundancy: the same airline and city names are repeated in every receipt, and renaming an airport means updating many rows.

Task 2. Normalization step by step

Step 1, 1NF. We split the lists so that each passenger with their seat gets a separate row. The key becomes composite because one booking now has several rows.

Booking_receipt(booking_id (PK), passenger_passport_number (PK), passenger_full_name, seat_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Step 2, 2NF. We move out columns that depend on only part of the key. Passenger data goes to its own table, booking data to another, and the seat stays in a linking table because it depends on both booking and passenger.

Passenger(passport_number (PK), full_name)

Booking(booking_id (PK), flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Booking_passenger(booking_id (PK, FK), passport_number (PK, FK), seat_number)

Step 3, 3NF. The Booking table still has transitive dependencies through `flight_number` and `airport_name`. We move flight, airport and airline data into separate tables and reference them by foreign keys.

Airline(airline_id (PK), airline_name)

Airport(airport_id (PK), airport_name, city)

Flight(flight_number (PK), departure_airport_id (FK → Airport), arrival_airport_id (FK → Airport), airline_id (FK → Airline))

Passenger(passport_number (PK), full_name)

Booking(booking_id (PK), flight_number (FK → Flight), ticket_price)

Booking_passenger(booking_id (PK, FK → Booking), passport_number (PK, FK → Passenger), seat_number)

Task 3. Functional dependencies

`passport_number` → `full_name`: one passport belongs to one person. Used for 2NF.

`booking_id` → `flight_number`, `ticket_price`: one booking is for one flight and has one price. Used for 2NF.

(`booking_id`, `passport_number`) → `seat_number`: a seat depends on both the booking and the passenger. Used for 2NF.

`flight_number` → `departure_airport`, `arrival_airport`, `airline_name`: a flight number defines its route and airline. Used for 3NF.

`airport_name` → `city`: an airport is always in one city. Used for 3NF.

Task 4. Is Flights in 3NF?

Yes, Flights is in 3NF.

Primary key: flight_id.

Functional dependency: flight_id → all other columns (times, airports, gates, airline, created_at, updated_at).

1NF: all values are atomic.

2NF: the key is one column, so partial dependencies are impossible.

3NF: there are no transitive dependencies. departing_airport_id, arriving_airport_id and airline_id are only foreign keys, and nothing else in Flights depends on them. The airport name, city and airline name are stored in the Airport and Airline tables. If Flights had a column like departing_airport_name, there would be a chain flight_id → departing_airport_id → departing_airport_name, which would violate 3NF. The other columns don't depend on each other either: an airport has many gates, so the gate is not determined by the airport, and actual times are not determined by scheduled times.

1. Creating tables

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, including the 'public' schema and its tables. The main query editor shows the following SQL code:

```
1 CREATE TABLE airline_info (
2     airline_id INT NOT NULL PRIMARY KEY,
3     airline_code VARCHAR(30) NOT NULL,
4     airline_name VARCHAR(50) NOT NULL,
5     airline_country VARCHAR(50) NOT NULL,
6     created_at TIMESTAMP NOT NULL,
7     updated_at TIMESTAMP NOT NULL,
8     info VARCHAR(50) NOT NULL
9 );
10
11 CREATE TABLE airport (
12     airport_id INT NOT NULL PRIMARY KEY,
13     airport_name VARCHAR(50) NOT NULL,
14     country VARCHAR(50) NOT NULL,
15     state VARCHAR(50) NOT NULL,
16     city VARCHAR(50) NOT NULL,
17     created_at TIMESTAMP NOT NULL,
18     updated_at TIMESTAMP NOT NULL
19 );
20
21 CREATE TABLE passengers (
22     passenger_id INT NOT NULL PRIMARY KEY,
```

The Data Output pane shows an error message: "ERROR: relation "airline_info" already exists". The SQL state is 42P07.

2. Check Not NULL

The screenshot shows the PostgreSQL IDE interface. The main query editor contains the following SQL code:

```
88 seat VARCHAR(50) NOT NULL,
89 boarding_time TIMESTAMP NOT NULL,
90 created_at TIMESTAMP NOT NULL,
91 updated_at TIMESTAMP NOT NULL
92 );
93
94 CREATE TABLE security_check (
95     security_check_id INT NOT NULL PRIMARY KEY,
96     check_result VARCHAR(20) NOT NULL,
97     created_at TIMESTAMP NOT NULL,
98     updated_at TIMESTAMP NOT NULL,
99     passenger_id INT NOT NULL
100 );
101
102
103
104
105 SELECT table_name, column_name, data_type, is_nullable
106 FROM information_schema.columns
107 WHERE table_schema = 'public'
108 ORDER BY table_name, ordinal_position;
```

The Data Output pane shows the results of the query, displaying a table with columns: table_name, column_name, data_type, and is_nullable. The results are as follows:

table_name	column_name	data_type	is_nullable
airline_info	airline_id	integer	NO
airline_info	airline_code	character varying	NO
airline_info	airline_name	character varying	NO
airline_info	airline_country	character varying	NO
airline_info	created_at	timestamp without time zone	NO
airline_info	updated_at	timestamp without time zone	NO
airline_info	info	character varying	NO
airport	airport_id	integer	NO
airport	airport_name	character varying	NO
airport	country	character varying	NO
airport	state	character varying	NO
airport	city	character varying	NO
airport	created_at	timestamp without time zone	NO
airport	updated_at	timestamp without time zone	NO
baggage	baggage_id	integer	NO

Total rows: 71. Query complete 00:00:00.106.

3. Changing table name

The screenshot shows the PostgreSQL GUI interface. On the left, the 'Tables (10)' folder under the 'public' schema is expanded, listing tables like 'airline', 'airport', 'baggage', etc. The main query editor on the right contains the following SQL code:

```
100 CREATE TABLE security_check (  
101     security_check_id INT NOT NULL PRIMARY KEY,  
102     check_result VARCHAR(20) NOT NULL,  
103     created_at TIMESTAMP NOT NULL,  
104     updated_at TIMESTAMP NOT NULL,  
105     passenger_id INT NOT NULL  
106 );  
107  
108 SELECT table_name, column_name, data_type, is_nullable  
109 FROM information_schema.columns  
110 WHERE table_schema = 'public'  
111 ORDER BY table_name, ordinal_position;  
112  
113 ALTER TABLE airline_info RENAME TO airline;  
114
```

The 'Data Output' tab shows the message: 'ALTER TABLE' and 'Query returned successfully in 58 msec.'

4. Changing price column

The screenshot shows the PostgreSQL GUI interface. On the left, the 'Columns (8)' folder under the 'booking' table is expanded, listing columns like 'booking_id', 'flight_id', 'passenger_id', etc. The main query editor on the right contains the following SQL code:

```
100 CREATE TABLE security_check (  
101     security_check_id INT NOT NULL PRIMARY KEY,  
102     check_result VARCHAR(20) NOT NULL,  
103     created_at TIMESTAMP NOT NULL,  
104     updated_at TIMESTAMP NOT NULL,  
105     passenger_id INT NOT NULL  
106 );  
107  
108 SELECT table_name, column_name, data_type, is_nullable  
109 FROM information_schema.columns  
110 WHERE table_schema = 'public'  
111 ORDER BY table_name, ordinal_position;  
112  
113 ALTER TABLE airline_info RENAME TO airline;  
114  
115 ALTER TABLE booking RENAME COLUMN price TO ticket_price;  
116
```

The 'Data Output' tab shows the message: 'ALTER TABLE' and 'Query returned successfully in 32 msec.'

5. Changing departing_gate

1.3 Sequences

Tables (10)

airline

airport

baggage

baggage_check

boarding_pass

booking

booking_flight

flights

Columns (12)

flight_id integer

sch_departure_time timestamp without time

sch_arrival_time timestamp without time zone

departing_airport_id integer

arriving_airport_id integer

departing_gate text

arriving_gate character varying(50)

airline_id integer

act_departure_time timestamp without time

act_arrival_time timestamp without time zone

created_at timestamp without time zone

updated_at timestamp without time zone

114

112

113

114

115

116

117

ALTER TABLE booking RENAME COLUMN price TO ticket_price;

ALTER TABLE flights ALTER COLUMN departing_gate TYPE TEXT;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 42 msec.

6. Dropping info column

Functions

Materialized Views

Operators

Procedures

1.3 Sequences

Tables (10)

airline

Columns (6)

airline_id integer

airline_code character varying(30)

airline_name character varying(50)

airline_country character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

Constraints

Indexes

RLS Policies

114

115

116

117

118

119

ALTER TABLE flights ALTER COLUMN departing_gate TYPE

ALTER TABLE airline DROP COLUMN info;

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 29 msec.

7. Checking

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

1.3 Sequences

Tables (10)

airline

Columns (6)

airline_id integer

airline_code character varying(30)

airline_name character varying(50)

airline_country character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

Constraints

Indexes

RLS Policies

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126

127

128

129

SELECT table_name, column_name, data_type

FROM information_schema.columns

WHERE table_schema = 'public'

AND (table_name = 'airline'

OR (table_name = 'booking' AND column_name = 'ticket_price')

OR (table_name = 'flights' AND column_name = 'departing_gate'))

ORDER BY table_name, ordinal_position;

Data Output

Messages

Notifications

Showing rows: 1 to 8

	table_name	column_name	data_type
1	airline	airline_id	integer
2	airline	airline_code	character varying
3	airline	airline_name	character varying
4	airline	airline_country	character varying
5	airline	created_at	timestamp without time zo...
6	airline	updated_at	timestamp without time zo...
7	booking	ticket_price	numeric
8	flights	departing_gate	text

8. Foreign keys

languages

publications

schemas (1)

> public

Aggregates

Collations

Domains

FTS Configurations

FTS Dictionaries

FTS Parsers

FTS Templates

Foreign Tables

Functions

Materialized Views

Operators

Procedures

Sequences

Tables (10)

airline

Columns (6)

airline_id integer

airline_code character varying(30)

airline_name character varying(50)

airline_country character varying(50)

created_at timestamp without time zone

updated_at timestamp without time zone

Constraints

Indexes

```

128 ORDER BY table_name, ordinal_position;
129
130
131 ALTER TABLE security_check ADD CONSTRAINT fk_security_check_passenger
132 FOREIGN KEY (passenger_id) REFERENCES passengers (passenger_id);
133
134 ALTER TABLE booking ADD CONSTRAINT fk_booking_passenger
135 FOREIGN KEY (passenger_id) REFERENCES passengers (passenger_id);
136
137 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_passenger
138 FOREIGN KEY (passenger_id) REFERENCES passengers (passenger_id);
139
140
141 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_booking
142 FOREIGN KEY (booking_id) REFERENCES booking (booking_id);
143
144 ALTER TABLE baggage ADD CONSTRAINT fk_baggage_booking
145 FOREIGN KEY (booking_id) REFERENCES booking (booking_id);
146
147 ALTER TABLE boarding_pass ADD CONSTRAINT fk_boarding_pass_booking
148 FOREIGN KEY (booking_id) REFERENCES booking (booking_id);

```

Data Output
Messages
Notifications

ALTER TABLE

Query returned successfully in 54 msec.

9. Check

```

170 SELECT tc.table_name      AS child_table,
171        kcu.column_name    AS child_column,
172        ccu.table_name     AS parent_table,
173        ccu.column_name    AS parent_column,
174        tc.constraint_name
175 FROM information_schema.table_constraints tc
176 JOIN information_schema.key_column_usage kcu
177 ON tc.constraint_name = kcu.constraint_name
178 AND tc.table_schema = kcu.table_schema
179 JOIN information_schema.constraint_column_usage ccu
180 ON tc.constraint_name = ccu.constraint_name
181 AND tc.table_schema = ccu.table_schema
182 WHERE tc.constraint_type = 'FOREIGN KEY'
183 AND tc.table_schema = 'public'
184 ORDER BY child_table, child_column;
185

```

Data Output
Messages
Notifications

Showing rows: 1 to 11

	child_table name	child_column name	parent_table name	parent_column name	constraint_name name
1	baggage	booking_id	booking	booking_id	fk_baggage_booking
2	baggage_ch...	booking_id	booking	booking_id	fk_baggage_check_booki...
3	baggage_ch...	passenger_id	passengers	passenger_id	fk_baggage_check_pass...
4	boarding_pa...	booking_id	booking	booking_id	fk_boarding_pass_booking
5	booking	passenger_id	passengers	passenger_id	fk_booking_passenger
6	booking_flight	booking_id	booking	booking_id	fk_booking_flight_booking
7	booking_flight	flight_id	flights	flight_id	fk_booking_flight_flight
8	flights	airline_id	airline	airline_id	fk_flights_airline
9	flights	arriving_airport_id	airport	airport_id	fk_flights_arriving_airport
10	flights	departing_airport...	airport	airport_id	fk_flights_departing_airp...
11	security_che...	passenger_id	passengers	passenger_id	fk_security_check_passe...

10. Booking receipt

```

186 CREATE SCHEMA IF NOT EXISTS norm;
187
188 CREATE TABLE norm.booking_receipt (
189     booking_id INT,
190     passenger_full_name VARCHAR(200),
191     passenger_passport_number VARCHAR(100),
192     flight_number VARCHAR(10),
193     departure_airport_name VARCHAR(100),
194     departure_city VARCHAR(50),
195     arrival_airport_name VARCHAR(100),
196     arrival_city VARCHAR(50),
197     airline_name VARCHAR(50),
198     seat_numbers VARCHAR(50),
199     ticket_price DECIMAL(9,2)
200 );
201
202 INSERT INTO norm.booking_receipt VALUES
203 (1001, 'Aruzhan Sadykova, Daniyar Sadykov', 'N12345601', 'N12345602', 'KC951',
204  'Almaty International Airport', 'Almaty',

```

booking_id	passenger_full_name	passenger_passport_number	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city
1	1001 Aruzhan Sadykova, Daniyar Sadykov	N12345601, N12345602	KC951	Almaty International Airport	Almaty	Nursultan Nazarbayev International Airp...	Astana
2	1002 Timur Akhmetov	N12345603	KC951	Almaty International Airport	Almaty	Nursultan Nazarbayev International Airp...	Astana
3	1003 Algerim Nurlanova	N12345604	TK351	Almaty International Airport	Almaty	Istanbul Airport	Istanbul
4	1004 Timur Akhmetov	N12345603	F21722	Almaty International Airport	Almaty	Dubai International Airport	Dubai
5	1005 Madina Karimova, Algerim Nurlan...	N12345605, N12345604	DV705	Nursultan Nazarbayev International Airp...	Astana	Almaty International Airport	Almaty
6	1006 Ertan Bekov	N12345606	TK351	Almaty International Airport	Almaty	Istanbul Airport	Istanbul
7	1007 Dana Serikova	N12345607	LH600	Frankfurt Airport	Frankfurt	Nursultan Nazarbayev International Airp...	Astana

11. Creating table

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 - Tables (10)
 - airline
 - Columns (6)
 - airline_id integer
 - airline_code character varying(30)
 - airline_name character varying(50)
 - airline_country character varying(50)
 - created_at timestamp without time zone
 - updated_at timestamp without time zone
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Query Query History

```

248     arrival_airport_id INT NOT NULL REFERENCES norm.airport (airport_id),
249     airline_id INT NOT NULL REFERENCES norm.airline (airline_id)
250 };
251
252 CREATE TABLE norm.passenger (
253     passport_number VARCHAR(20) NOT NULL PRIMARY KEY,
254     full_name VARCHAR(100) NOT NULL
255 );
256
257 CREATE TABLE norm.booking (
258     booking_id INT NOT NULL PRIMARY KEY,
259     flight_number VARCHAR(10) NOT NULL REFERENCES norm.flight (flight_number),
260     ticket_price DECIMAL(9,2) NOT NULL
261 );
262
263 CREATE TABLE norm.booking_passenger (
264     booking_id INT NOT NULL REFERENCES norm.booking (booking_id),
265     passport_number VARCHAR(20) NOT NULL REFERENCES norm.passenger (passport_number),
266     seat_number VARCHAR(5) NOT NULL,
267     PRIMARY KEY (booking_id, passport_number)
268 );
269

```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 52 msec.

12. Insert

```

299 ('N12345607', 'Dana Serikova');
300
301 INSERT INTO norm.booking (booking_id, flight_number, ticket_price) VALUES
302 (1001, 'KC951', 70000.00),
303 (1002, 'KC951', 35000.00),
304 (1003, 'TK351', 180000.00),
305 (1004, 'FZ1722', 120000.00),
306 (1005, 'DV705', 60000.00),
307 (1006, 'TK351', 175000.00),
308 (1007, 'LH600', 250000.00);
309
310 INSERT INTO norm.booking_passenger (booking_id, passport_number, seat_number) VALUES
311 (1001, 'N12345601', '12A'),
312 (1001, 'N12345602', '12B'),
313 (1002, 'N12345603', '14C'),
314 (1003, 'N12345604', '7F'),
315 (1004, 'N12345603', '21A'),
316 (1005, 'N12345605', '3A'),
317 (1005, 'N12345604', '3B'),
318 (1006, 'N12345606', '9D'),
319 (1007, 'N12345607', '2A');
320

```

Data Output Messages Notifications

INSERT 0 9

Query returned successfully in 39 msec.

30)

(50)

ng(50)

time zon

time zor

13. Check

```

321
322 SELECT b.booking_id,
323        p.full_name      AS passenger_full_name,
324        p.passport_number AS passenger_passport_number,
325        f.flight_number,
326        da.airport_name  AS departure_airport_name,
327        da.city          AS departure_city,
328        aa.airport_name  AS arrival_airport_name,
329        aa.city          AS arrival_city,
330        al.airline_name,
331        bp.seat_number,
332        b.ticket_price
333 FROM norm.booking b
334 JOIN norm.booking_passenger bp ON bp.booking_id = b.booking_id
335 JOIN norm.passenger p         ON p.passport_number = bp.passport_number
336 JOIN norm.flight f           ON f.flight_number = b.flight_number
337 JOIN norm.airport da         ON da.airport_id = f.departure_airport_id
338 JOIN norm.airport aa         ON aa.airport_id = f.arrival_airport_id
339 JOIN norm.airline al         ON al.airline_id = f.airline_id
340 ORDER BY b.booking_id, bp.seat_number;
341

```

Data Output Messages Notifications

Showing rows: 1 to 9 of 1 Page No: 1

	booking_id integer	passenger_full_name character varying (100)	passenger_passport_number character varying (20)	flight_number character varying (10)	departure_airport_name character varying (100)	departure_city character varying (50)	arrival_airport_name character varying (100)	arrival_city character varying (50)
1	1001	Aruzhan Sadykova	N12345601	KC951	Almaty International Airport	Almaty	Nursultan Nazarbayev International Airp...	Astana
2	1001	Daniyar Sadykov	N12345602	KC951	Almaty International Airport	Almaty	Nursultan Nazarbayev International Airp...	Astana
3	1002	Timur Akhmetov	N12345603	KC951	Almaty International Airport	Almaty	Nursultan Nazarbayev International Airp...	Astana
4	1003	Aigerim Nurlanova	N12345604	TK351	Almaty International Airport	Almaty	Istanbul Airport	Istanbul
5	1004	Timur Akhmetov	N12345603	FZ1722	Almaty International Airport	Almaty	Dubai International Airport	Dubai
6	1005	Madina Karimova	N12345605	DV705	Nursultan Nazarbayev International Airp...	Astana	Almaty International Airport	Almaty
7	1005	Aigerim Nurlanova	N12345604	DV705	Nursultan Nazarbayev International Airp...	Astana	Almaty International Airport	Almaty
8	1006	Erlan Bekov	N12345606	TK351	Almaty International Airport	Almaty	Istanbul Airport	Istanbul
9	1007	Dana Serikova	N12345607	LH600	Frankfurt Airport	Frankfurt	Nursultan Nazarbayev International Airp...	Astana